Student 2 · Master File

Everything you need in one place: the study, your Results part from zero, your slides, your questions.
Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad

> **Your role.** You present slides 13–23 (Results, about 6 minutes) and you are the team's statistics person. You own Chapter 3 (Results), Tables 1–13, Figures 3–11 and Appendix A, and you must understand Section 2.6 (Statistical analysis). Handover line at the end of your part: “Those are our results. My colleague will now discuss what they mean.”

Contents

1. **The study in full (aim, hypotheses, methods, conclusions)**
2. **Whole-study essentials: two-minute script, core facts, terms**
3. **The Results as one story**
4. **Statistics from zero (with our numbers)**
5. **Every table and figure: why it is there and how to read it**
6. **Your chapter, section by section**
7. **Statistical analysis (Section 2.6, word for word)**
8. **Your slides: what to point at, and the full scripts**
9. **Questions and model answers, plus drills**
10. **Your key sources**
11. **Editing your chapter**
12. **Your 14-day plan, self-test and day-before checklist**
13. **Cheat sheet**

How to study this file

Day	Read
1	Section 1 (the study) and Section 2 (essentials). Say the two-minute script out loud three times.
2	Section 3 (the story) and Section 4, lessons 1–12.
3	Section 4, lessons 13–25. Explain each lesson out loud using our numbers.
4–5	Section 5 (every table and figure), with the thesis open beside you.
6	Sections 6 and 7 (your chapter and the methods).
7–8	Section 8: practise your slides alone, three times, with a timer.
9–10	Section 9: answer every question out loud; record yourself once.
11–14	Sections 10–12; rehearsals with the team; self-test until 100%; Section 13 in your pocket.

1 · The study in full

Title

Health Anxiety, AI Chatbot Use and Cyberchondria Among Adults in Baghdad: An Online Cross-Sectional Survey

Abstract (as in the thesis)

Background: Adults increasingly consult search engines and, more recently, artificial-intelligence (AI) chatbots before seeing a doctor. Health anxiety and cyberchondria (excessive, distressing online health searching) are thought to reinforce each other, but evidence from Iraq, and on AI chatbots, is scarce.

Objectives: To measure the association between health anxiety and cyberchondria among adults in Baghdad, and to describe online and AI health-information seeking and pre-consultation behaviour.

Methods: Cross-sectional online survey of 300 adults in Baghdad (July–September 2026) using the Short Health Anxiety Inventory (SHAI-14) and the Cyberchondria Severity Scale (CSS-12). Analyses included measurement-invariance testing across Arabic and English responders, FDR-corrected group comparisons, robust regression and bootstrapped mediation.

Results: Participants were mostly young (62.3% aged 20–29) and 63.3% had a health-related background; the CSS-12 was invariant across Arabic and English responders. AI chatbots were ticked by 78.0% (88.7% including respondents who named a chatbot), against 56.3% for search engines. Health anxiety correlated with cyberchondria ($r = 0.47$, 95% CI 0.38–0.55) and remained its strongest predictor after adjustment ($\beta = 0.48$), most strongly with excessive searching and post-search distress. Elevated health anxiety (SHAI-14 ≥ 18, a screening threshold rather than a diagnosis) was found in 19.0%; only 0.7% reached the stricter low-prevalence cut-off of 29. After searching, 34.7% delayed or skipped care because they felt reassured and 31.7% never told their doctor, citing fear of judgement; higher cyberchondria was associated with more disclosure. The association between AI use and cyberchondria was not robust to how AI use was defined.

Conclusion: In this convenience sample of mostly young, health-oriented adults, health anxiety and cyberchondria were linked, and AI chatbots were a major channel for online health information, without reliable evidence that their use adds to cyberchondria beyond health anxiety. The cross-sectional design cannot establish direction. Clinicians should ask about online and AI searches without judgement.

Keywords: cyberchondria; health anxiety; artificial intelligence; chatbots; online health information seeking; Iraq.

Why the study was done (Rationale, 1.7)

This study addresses three gaps. First, it provides, to our knowledge, the first estimate of the association between health anxiety and cyberchondria (using the CSS-12) in a Baghdad-based sample. Second, it examines AI chatbots as a distinct, and now dominant, source of online health information, rather than treating "the Internet" as one entity. Third, it links these psychological constructs to concrete pre-consultation behaviours, namely delaying care, self-medication, urgent visits, multiple consultations, disclosure of searches and trust in physicians, that matter directly to clinicians.

Aim, objectives and hypotheses (1.8, word for word)

Aim: To study the relationship between health anxiety and cyberchondria and their correlates among adults in Baghdad who search for health information online, with particular attention to AI chatbots and pre-consultation behaviour.

Specific objectives:

1. To describe online and AI-based health-information seeking and pre-consultation behaviour among adults in Baghdad.

2. To estimate levels of health anxiety and cyberchondria and the strength of their association, overall and by cyberchondria dimension, independently of socio-demographic and digital-use factors (H1, H2); correlates of each construct were examined as secondary analyses.

3. To examine statistical pathways consistent with a mutually reinforcing cycle, and factors that might modify the association (H3).

4. To examine how health anxiety and cyberchondria relate to pre-consultation behaviour, disclosure of searches to physicians and trust in physicians (H4).

Hypotheses: (H1) Health anxiety is positively associated with cyberchondria, most strongly with its distress and excessiveness dimensions. (H2) The association is independent of socio-demographic and digital-use factors. (H3) Excessive searching statistically mediates the link between health anxiety and post-search distress. (H4) Higher cyberchondria is associated with care-seeking behaviours (urgent visits, multiple consultations) and with a more strained doctor–patient relationship. H1 and H2 were the primary hypotheses and H3 and H4 secondary; all other analyses, including the AI-use comparisons, item-level, moderation and sensitivity analyses, were exploratory.

The conceptual framework of the study, linking these hypotheses to the proposed cycle between health anxiety and the facets of cyberchondria, is shown in Figure 1.

Were the hypotheses supported? (memorise this table)

Hypothesis	Verdict	Evidence (where)
H1: health anxiety is positively associated with cyberchondria, most with distress and excessiveness	Supported	r = 0.47 (0.38–0.55); excessiveness 0.46 and distress 0.45 strongest (Table 6, Figure 5)
H2: independent of socio-demographic and digital-use factors	Supported	partial r = 0.48; health anxiety adds 18.4%, β 0.48 (Table 9)
H3: excessive searching mediates health anxiety → post-search distress	Supported statistically, not causally	indirect 0.16 (0.11–0.23), but reverse path also 0.15 (Table 12, Figure 9)
H4: cyberchondria ↔ care-seeking and strained relationship	Partially supported	only more disclosure survived FDR (AOR 0.57, q 0.012); urgent visits, strained relationship exploratory (Table 11)

> **Say:** *H1 and H2 were supported; H3 was supported statistically but not causally; H4 was only partially supported.*

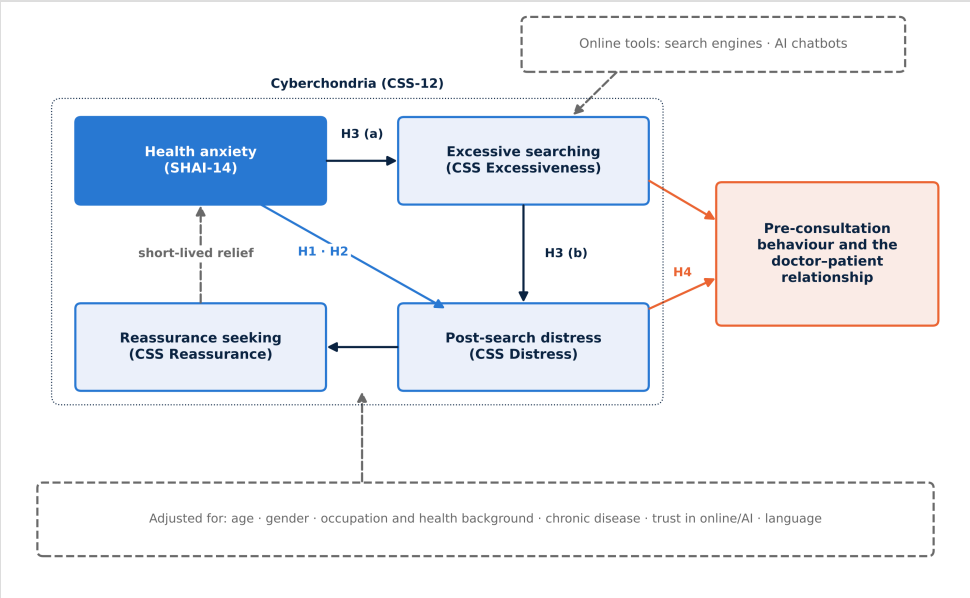


Figure 1: conceptual framework (where each hypothesis sits).

Methods (2.1–2.5, word for word; 2.6 is in Section 7)

This was an analytical cross-sectional study reported in line with the STROBE statement for observational studies [37] and the CHERRIES checklist for web-based surveys [38]. Data were collected online in Baghdad, Iraq, between 17 July 2026 and 11 September 2026.

2.2 Participants and sampling

A non-probability (convenience and snowball) sampling approach was used. The survey link was distributed through online channels (social media and messaging groups) whose members were based in Baghdad. Inclusion criteria were: age 18 years or older, ability to read Arabic or English, and provision of informed consent. Every completed submission was retained for analysis. No duplicate response patterns were found. [Authors to state whether Google Forms limited submissions to one per account.] Because the link spread by snowball sampling, the number of people who saw the invitation is unknown, so view and participation rates could not be computed. Three responses had been annotated by the team during data collection [authors to state the reason for each annotation]; they were kept, and their exclusion was tested in a sensitivity analysis. The resulting sample was predominantly young and from health-related fields (Section 3.1).

2.3 Sample size

The study was powered for its primary objective, the correlation between health anxiety and cyberchondria. Using Fisher's z approximation with two-sided $\alpha = 0.05$ and 80% power, a minimum of 194 participants is required to detect a small-to-moderate correlation ($r = 0.20$), and 85 to detect $r = 0.30$ [39]. Meta-analytic estimates of this association are far larger ($r \approx 0.62$) [18,19], so these targets are conservative. For descriptive proportions, 300 participants give a margin of error of ±5.7% at 95% confidence (worst case, $p = 0.5$), compared with ±5% for the 385 participants of a Cochran-based calculation. The logistic model for elevated health anxiety had 57 events for 11 predictors (≈5 events per variable), which is acknowledged as a limitation of that model.

2.4 Data collection instrument

A bilingual (Arabic–English) self-administered questionnaire was built in Google Forms. All items were required, except the free-text description of chronic disease and the name of the AI chatbot used, which were conditional. The questionnaire had five parts:

1. Consent and socio-demographic data: confirmation of age ≥18 and consent; age group; gender; highest education; nature of study or work; diagnosed chronic medical condition (and its type).

2. Digital health information seeking: tools used (search engines, AI chatbots, dedicated medical platforms, social media; multiple answers); main AI chatbot; motivation for searching (multiple answers); and trust in medical information generated by online tools/AI (5-point scale).

3. Health anxiety: the 14-item Short Health Anxiety Inventory (SHAI-14) [40]. Each item offers four statements scored 0–3, giving a total of 0–42; higher scores indicate greater health anxiety. The SHAI has good reliability and validity in clinical and non-clinical samples, including recent validations in Spain, Sweden and Iran [41–43], and an Arabic version has been validated in Saudi Arabia [44].

4. Cyberchondria: the CSS-12 [14], with items rated on a 5-point frequency scale (1 = never to 5 = always), giving a total of 12–60 and four subscales (3–15 each): Excessiveness (items 1–3), Compulsion (4–6), Distress (7–9) and Reassurance seeking (10–12).

5. Pre-consultation behaviour and relationship with physicians: behavioural responses after searching (multiple answers), whether search or AI results are disclosed to the physician, the perceived effect of digital information seeking on the relationship with and trust in physicians, and the language mainly relied upon while answering.

Each item was presented in English and Arabic. The research team produced the Arabic wording; its equivalence was examined empirically by comparing the psychometric behaviour of the scales between participants who relied mainly on Arabic and those who relied mainly on English (Section 2.6). For one SHAI item (item 2, noticing aches and pains) the four statements were displayed in a different order from the original scale ("as much as most people" before "less than most people"); all other SHAI and CSS items appeared in their original order. Items were scored according to the meaning of the chosen statement, following the original scoring key. The Arabic wording was produced by the research team rather than through formal cross-cultural adaptation [45], and the published Arabic versions of the CSS-12 and SHAI [15,44,46] were not used. As in the original scale, CSS item 3 refers to entering symptoms into a search engine.

2.5 Variables and operational definitions

Elevated health anxiety: SHAI-14 ≥ 18, the optimal cut-off for distinguishing people with health anxiety from healthy controls in a Swedish diagnostic-accuracy study [42]. Because the same study recommends higher cut-offs in psychiatric (22) and low-prevalence (29) settings, the term "elevated" rather than "clinically relevant" is used, it is not a diagnosis, and prevalence at 15, 22 and 29 is also reported.

High cyberchondria: CSS-12 total in the upper tertile of the sample (> 33), since the CSS-12 has no validated clinical cut-off.

AI-chatbot user: ticked "AI chatbots" as a tool or named an AI chatbot in the follow-up question. Thirty-two respondents named a chatbot without ticking the tool option; they were classified as users (the follow-up question was optional and specifically addressed to chatbot users). Because this decision affects the results, AI use is reported both ways (ticked only, and ticked or named), and all analyses involving AI use were repeated with the stricter definition.

Health-related background: medical students, students in other health fields and healthcare workers.

Occupation: free-text "Other" answers were recoded into the closest category (e.g. secondary-school students and school leavers as non-health students; housewives, engineers and business owners as non-health employed/other). Age groups 40–49, 50–59 and 60+ were merged (≥40) for inferential analysis because of small cell sizes.

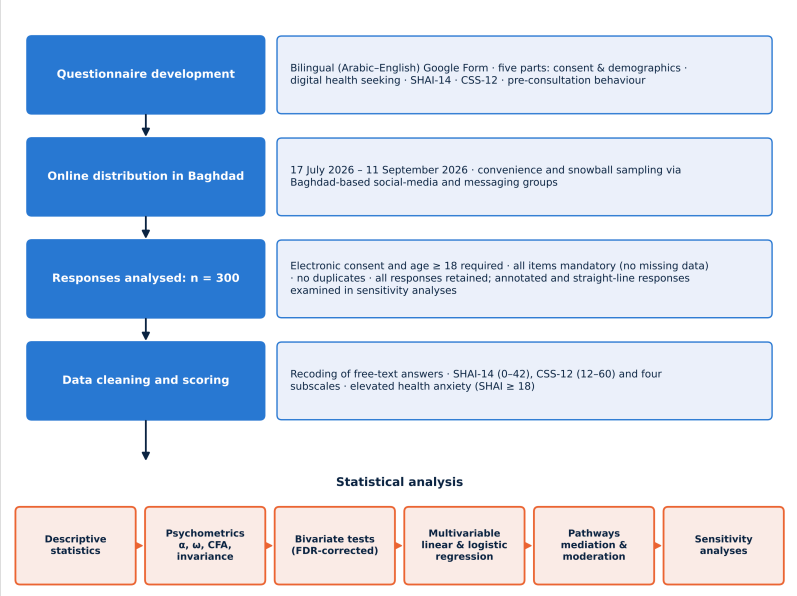


Figure 2: study flowchart.

Ethics (2.7)

Participation was voluntary and anonymous. The first page of the form explained the purpose of the study, and electronic informed consent (with confirmation of age ≥18) was required before any question was shown; participants could withdraw at any time by closing the form. No names or contact details were collected, and data were stored securely and used only for research purposes. [Ethical approval: committee name, approval number and date to be inserted.]

Conclusions (5.1, word for word)

1. AI chatbots, mainly ChatGPT, have become a leading source of health information before consulting a doctor among digitally active adults in Baghdad: 78.0% ticked them (88.7% including those who named a chatbot), more than used search engines.

2. About one in five participants (19%) had elevated health anxiety (SHAI ≥ 18, a screening threshold rather than a diagnosis; 0.7% at the stricter cut-off of 29), and younger people and those searching for instant reassurance were most affected.

3. Health anxiety and cyberchondria are moderately and robustly associated (r = 0.47); health anxiety is the strongest independent predictor of cyberchondria, especially of excessive searching and post-search distress.

4. The pattern of associations is compatible with a reinforcing cycle (health anxiety, excessive searching, distress, reassurance-seeking consultations), but the cross-sectional decomposition, based on subscales of one instrument, cannot establish direction.

5. There is no reliable evidence that AI-chatbot use is associated with cyberchondria independently of health anxiety: the association depended on how inconsistent answers about AI use were classified, and AI use did not change the strength of the health anxiety–cyberchondria link.

6. Online searching shapes the consultation: about one-third of participants delay care because they feel reassured, about one-third never disclose their searches to their doctor for fear of judgement, and about one-third say searching has complicated or reduced their trust in physicians. Higher cyberchondria was robustly linked with more disclosure; links with urgent care-seeking, concealment by health-anxious participants and a strained relationship were exploratory.

Recommendations (5.2, word for word)

For clinicians and medical education:

Routinely and non-judgementally ask patients what they have searched for or asked an AI chatbot, to bring hidden information into the consultation and reduce concealment.

Recognise repeated searching, post-search distress and repeated reassurance seeking as possible markers of health anxiety, and screen with a brief tool such as the SHAI where appropriate; offer cognitive-behavioural strategies rather than repeated reassurance or investigations.

Include digital health communication, AI literacy and the management of health anxiety in undergraduate medical curricula, and support medical students who may themselves be vulnerable.

For the Ministry of Health and public-health bodies:

The Ministry of Health may consider campaigns on digital and AI health literacy that explain both the uses and the limits of online information, especially false reassurance that leads to delayed care.

It may also consider providing or endorsing reliable Arabic-language health information sources and triage services so that people searching for "whether I need a doctor" are directed to safe advice.

For developers of AI tools:

Developers may consider designing health responses that recognise signs of health anxiety (e.g. repeated questions about the same symptom), avoid escalating rare diagnoses without context, and encourage appropriate professional consultation, in Arabic as well as English.

For future research:

Develop and test interventions: ClinicalTrials.gov lists only nine registered studies on cyberchondria (October 2026), none in Iraq and six of them observational and three interventional [60], so trials of brief digital health-literacy or cognitive-behavioural interventions in Arabic are a clear gap.

Conduct longitudinal or experience-sampling studies to test whether health anxiety and cyberchondria reinforce each other over time, and experimental studies comparing AI chatbot and search-engine answers on post-search anxiety.

Use probability sampling of the general Baghdad population, including older and less-educated adults, and use formally adapted Arabic instruments (the published CSS-12-Ar and Arabic SHAI, or a new forward–backward adaptation with measurement-invariance testing), with CSS item 3 worded to cover both search engines and AI chatbots.

Link self-reported behaviours to objective healthcare utilisation and clinical outcomes, especially delayed diagnosis following false reassurance.

Strengths and limitations (4.8, word for word)

Strengths of this study include the use of two established instruments with good reliability and confirmed structure; a bilingual questionnaire with formal tests of measurement invariance across languages; the separate assessment of AI chatbots; complete data; correction for multiple testing; robust regression, bootstrap inference and several sensitivity analyses; and the linking of psychological measures to concrete behaviours relevant to clinical practice.

The limitations are substantial and should be weighed carefully:

Design and causality: the cross-sectional design cannot establish temporal order; mediation results describe statistical pathways, not causal mechanisms, and may be biased relative to longitudinal effects.

Sampling and generalisability: an online convenience/snowball sample is subject to self-selection. The sample was young, highly educated and over-represented medical students (44.0%) and people with health backgrounds (63.3%), so prevalence estimates and the frequency of AI use cannot be generalised to the adult population of Baghdad. Residence in Baghdad was assured by the distribution channels rather than verified by a question, and non-Internet users, older adults and less-educated people are under-represented by design.

Measurement: the Arabic wording was produced by the research team without a formal forward–backward translation and cognitive pre-testing; formal tests supported full (scalar) invariance of the CSS-12 and metric invariance of the SHAI-14 across language groups, with no item showing DIF after correction, but SHAI-14 intercepts were not fully equivalent and its baseline model fitted poorly, so its invariance results are hard to interpret, the language groups were self-selected rather than randomised, and statistical invariance does not replace formal cross-cultural adaptation with cognitive pre-testing. Published Arabic versions of both scales exist [15,44,46] but were not used, so comparability with Arabic studies that used them is uncertain. On the form, the response options of one SHAI item (item 2) were displayed out of their original order; scoring followed the original key, and dropping the item did not change the main result, but the order may still have influenced answers. CSS item 3 refers only to search engines and may under-capture repeated querying of AI chatbots. "Language relied on" was self-selected, not randomised. The SHAI-14 one-factor model fitted poorly under maximum-likelihood estimation. The SHAI cut-off was derived in Sweden, and the CSS-12 has no validated clinical cut-off, so the "high cyberchondria" group is sample-relative.

Single-item and self-designed measures: trust, motivations, behaviours, disclosure and relationship were assessed with single, non-validated items, some allowing multiple answers. The questionnaire asked for "most frequent" behaviours but allowed several answers; the trust item combined online tools and AI, so it cannot separate trust in search results from trust in chatbots; the disclosure option combined frequency with a reason (fear of judgement); and "somewhat complicated" is not clearly a negative effect on the relationship.

Self-report bias: all measures were self-reported and may be affected by recall and social desirability bias; no clinical data or diagnoses were available, and chronic disease was self-reported.

Multiple testing and power: despite FDR correction, the large number of exploratory tests means that some isolated associations may be false positives; several behavioural associations were nominally significant only. Some subgroups (non-AI users, n = 34; healthcare workers, n = 15; older age groups) were small, and the logistic model for elevated health anxiety had about five events per predictor.

Classification of AI use: 32 respondents named an AI chatbot without ticking the AI option. They were classed as users, and the independent association between AI use and cyberchondria depended on this decision, so it should be regarded as tentative.

Data quality: an online survey cannot exclude careless responding entirely; nine straight-line responders were identified, although their exclusion did not change the results.

2 · Whole-study essentials

> **The committee can ask any of you about any part.** You split the presentation, not the knowledge. Learn this part so well that you can explain the whole study without slides.

The study in two minutes (practise saying this out loud)

"Many people now look up their symptoms online before seeing a doctor, and more and more of them ask AI chatbots like ChatGPT. For some people this searching makes them more worried, not less. That is called cyberchondria, and it is closely linked to health anxiety. No one had studied this in Baghdad, and AI chatbots had hardly been studied next to search engines. We ran an online survey of 300 adults in Baghdad between 17 July 2026 and 11 September 2026, using two established scales: the SHAI-14 for health anxiety and the CSS-12 for cyberchondria. We found that AI chatbots are now the main source of health information: 78.0% ticked them, more than search engines (56.3%). About one in five (19%) had elevated health anxiety. Health anxiety and cyberchondria were moderately linked, r = 0.47, and this held after adjusting for everything else. The link ran mainly through excessive searching and post-search distress, which fits a vicious cycle, although one survey cannot prove direction. We found no reliable evidence that AI itself adds to cyberchondria beyond health anxiety. And searching shapes the consultation: about a third delay care because they feel reassured, and about a third never tell their doctor what they searched. So our main message for clinicians is simple: ask patients, without judgement, what they searched or asked an AI."

Core facts (memorise; examiners test these)

Topic	Fact
Design	Analytical cross-sectional online survey; STROBE and CHERRIES checklists
Who and when	**300** adults, Baghdad, 17 July 2026 – 11 September 2026; convenience + snowball sampling
Sample size	**194** needed for r = 0.20 (α 0.05, 80% power); margin of error ±5.7%
Sample profile	62.3% aged 20–29 · 56.3% female · **44.0%** medical students · 63.3% health background · 21.7% chronic condition · 70.7% answered in Arabic
Tools used	AI chatbots **78.0%** ticked (88.7% incl. named) · search engines 56.3% · medical platforms 18.3% · social media 10.7% · ChatGPT 59.3% · AI without any search engine 41.0%
Motives	convenience 80.0% · "do I need a doctor?" 57.7% · instant reassurance 51.0%
Health anxiety	SHAI mean 12.3 ± 5.8 (0–42) · **19%** elevated (≥ 18; 95% CI 15.0–23.8%) · 0.7% at ≥ 29
Cyberchondria	CSS mean 28.9 ± 8.4 (12–60) · excessiveness highest, compulsion lowest
Reliability	α 0.81 (SHAI), 0.86 (CSS); ω 0.82, 0.87 · CSS-12 4-factor CFA: CFI 0.95, RMSEA 0.070, SRMR 0.053
Arabic vs English	CSS-12 fully (scalar) invariant; SHAI-14 metric invariant; no item DIF after FDR
Main result	**r = 0.47** (95% CI 0.38–0.55); explains 22% of the variance · partial r = 0.48
By facet	excessiveness 0.46 · distress 0.45 · compulsion 0.25 · reassurance 0.24
Regression	R^2 0.030 (demographics) → 0.133 (+ digital use) → 0.317 (+ health anxiety, ΔR^2 18.4%); β = 0.48
Who is anxious	AOR **3.12** per SD of cyberchondria · 3.83 reassurance motive · 0.61 per older age group · AUC 0.83 → 0.79
Pathways	via excessive searching 0.16 (CI 0.11–0.23) · via distress 0.22 (direct → 0.01) · reverse 0.15
Behaviour	delay care when reassured **34.7%** · ignore symptom 30.0% · several doctors 25.3% · urgent care 22.0% · self-medication 21.3%

Doctor relationship	never tell the doctor **31.7%** · always tell 27.0% · trust strained 34.7% · higher cyberchondria → more disclosure (OR 0.57, q 0.012)
AI effect	+5.3 CSS points (inclusive) but +1.5, p = 0.31 (strict) → no reliable independent effect · 32 named-not-ticked · 34 non-users
Robustness	every sensitivity analysis left the main result unchanged (Table 13)

Terms you must be able to explain in one sentence

Term	Plain explanation
Health anxiety	Persistent worry about having or developing a serious illness, kept going by misreading normal body sensations and by reassurance seeking.
Illness anxiety disorder	The DSM-5 diagnosis at the severe end of health anxiety. We did **not** diagnose it; we measured symptoms.
Cyberchondria	Excessive or repeated online health searching that increases rather than relieves distress.
SHAI-14	14 items, each scored 0–3 (total 0–42); higher = more health anxiety. ≥ 18 = elevated (screening threshold).
CSS-12	12 items scored 1–5 (total 12–60) with four 3-item facets: excessiveness, compulsion, distress, reassurance seeking.
Cross-sectional	Everything measured once, at one time point: shows associations, not cause and effect.
Convenience / snowball sampling	People who were easy to reach, who then shared the link with others. Quick, but not representative.
Mean ± SD	The average score and how spread out the scores are.
Median (IQR)	The middle score and the range of the middle half; used when data are skewed.
p-value	How surprising the result would be if there were truly no effect; $p < 0.05$ is called significant.
95% CI	The range in which the true value probably lies; narrower = more precise.
Correlation r	How closely two scores move together (0 = none, 1 = perfect). 0.47 is moderate.
Partial correlation	The same link after removing the effect of other variables.
Mann–Whitney / Kruskal–Wallis	Non-parametric tests comparing two / three or more groups, used because scores were not normally distributed.
χ^2 test	Tests whether two categorical variables are related (e.g., social-media use and elevated health anxiety).
FDR (q-value)	Correction for running many tests; a result is “robust” if $q < 0.05$, otherwise exploratory.
Linear regression, B and β	Predicts a score from several factors at once. B = change in CSS points per unit; β = the same on a standardised scale (to compare factors).
R^2 and ΔR^2	Share of variation explained by the model; ΔR^2 = extra share explained by adding a block of variables.
Logistic regression, AOR	Predicts a yes/no outcome. Adjusted odds ratio: how many times the odds change, holding other factors constant.
AUC	How well a model separates two groups (0.5 = chance, 1 = perfect).
Mediation	Splitting a link into a part that runs through a third variable (indirect) and the rest (direct). Statistical, not causal.
Moderation	Whether a third variable changes the strength of a link (e.g., does AI use make the link stronger? No).

Cronbach α / McDonald ω	Internal consistency: do the items of a scale hang together? ≥ 0.7 is acceptable, ≥ 0.8 good.
CFA (CFI, RMSEA, SRMR)	Checks the items group into the intended factors. Good: CFI ≈ 0.95+, RMSEA ≤ 0.08, SRMR ≤ 0.08.
Measurement invariance	Checks the scale measures the same thing in two groups (here Arabic vs English) before comparing them.
Sensitivity analysis	Re-running the main analysis in a different way to see if the result holds.

Never say (overclaims the committee will catch):

- "Health anxiety *causes* cyberchondria" → say "is associated with", "is compatible with a cycle".
- "19% *have* health anxiety disorder" → "19% scored in the elevated range on a screening tool".
- "AI makes people anxious" → "we found no reliable independent effect of AI use".
- "Our results represent Baghdad" → "a young, online, convenience sample; associations were consistent across subgroups".
- "The Arabic version is validated" → "we tested it: equal reliability and formal invariance, but no formal back-translation".

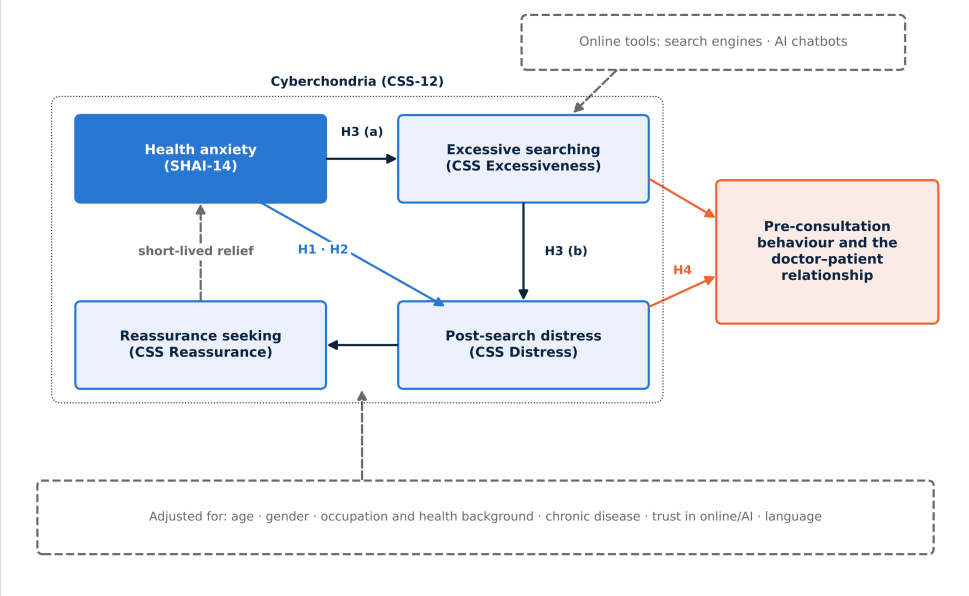


Figure 1 of the thesis: the conceptual framework. Learn to explain every arrow.

3 · The Results as one story

Every table and figure in Chapter 3 answers one question. They come in a logical order: first **who** answered, then **what they do**, then **how anxious** they are, then the **main link**, then **what else** could explain it, then **who is at risk**, then **behaviour**, then **pathways**, and finally **is it robust?**

Step	Question	Short answer	Where	Slide
1	Who took part?	300 young, educated adults; 44.0% medical students	Table 1	14
2	How do they search?	AI chatbots first (78.0%), search engines second (56.3%)	Table 2, Figure 3	15
3	What do they do after searching?	34.7% delay care when reassured; 31.7% never tell the doctor	Figure 4	22
4	How anxious are they? Do the scales work?	19% elevated; both scales reliable; CSS-12 same in Arabic and English	Tables 3, 4, 5	16
5	Are health anxiety and cyberchondria linked? (H1)	Yes, r = 0.47	Table 6, Figures 5, 6, 11	17, 19
6	Which groups score higher?	Chronic illness, reassurance seekers; AI users (but see step 7)	Tables 7, 8, Figure 7	–
7	Does the link survive other factors? (H2) Who is at risk?	Yes; health anxiety adds 18.4%; AI effect not robust	Tables 9, 10, Figure 8	18, 20, 23
8	Does it affect behaviour? (H4)	Cyberchondria → more disclosure (robust); others exploratory	Table 11	22
9	Is there a cycle? (H3) Is everything robust?	Compatible with a cycle, not proven; all checks unchanged	Table 12, Figures 9, 10, Table 13	21

> **Say:** *Our results answer nine questions in order. In short: AI chatbots are the main source of health information, one in five has elevated health anxiety, and health anxiety and cyberchondria are clearly linked, r = 0.47, even after adjusting for everything else.*

4 · Statistics from zero (with our numbers)

Each lesson: the idea in plain words, our example, and what to say. Read them in order; each one builds on the one before.

1. Variables: two kinds

- **Categorical** = groups (gender, uses AI yes/no, tool profile). We describe them with **n and %**.
- **Continuous** = scores on a scale (SHAI 0–42, CSS 12–60). We describe them with **mean, SD, median**.

This one fact decides almost every choice of test and chart in the thesis.

2. Mean and SD

Mean = the average. **SD (standard deviation)** = how spread out people are around the average. About two-thirds of people lie within 1 SD of the mean.

```
SHAI mean 12.3 ± SD 5.8 → most people scored between about 6.5 and 18
```

> **Say:** *The average health-anxiety score was 12.3 out of 42.*

3. Median and IQR

The **median** is the middle person. The **IQR** is the range of the middle half. We use them when scores are skewed (not a symmetrical bell shape).

```
SHAI median 12 (IQR 8–16): half of people scored 8 to 16
```

4. Normal distribution and why it matters

Many tests assume scores form a bell curve. The **Shapiro–Wilk test** checks this. Our scores were slightly skewed ($p < 0.01$), so we used **non-parametric** tests (they do not need the bell shape) and **robust** regression.

> **Say:** *Our scores were not perfectly normal, so we used tests that do not assume normality.*

5. p-value

Imagine there is truly **no** link. The p-value is the chance of seeing a result as strong as ours just by luck. Small p (below 0.05) = "unlikely to be luck" = **significant**. It does **not** say how big or important the effect is.

```
r = 0.47, p < 0.001 → almost impossible to get this by chance if there were no
link
```

> **Trap:** p tells you if an effect exists, not how big it is. For size, look at r, β, OR.

6. 95% confidence interval (CI)

Our number is an estimate from 300 people. The CI is the range where the true value probably lies. Narrow CI = precise. If a CI for an odds ratio includes 1, the result is not significant.

```
r = 0.47, 95% CI 0.38–0.55 → the true correlation is probably between 0.38 and 0.55: always clearly above zero
```

7. Correlation r

How closely two scores move together: 0 = no link, 0.1 small, 0.3 moderate, 0.5 large, 1 = perfect. Positive = both go up together.

```
r = 0.47: people with more health anxiety tend to have more cyberchondria. r² = 22%: health anxiety explains about a fifth of the differences in cyberchondria.
```

> **Say:** *Health anxiety and cyberchondria rise together: r = 0.47, a moderate link.*

8. Pearson vs Spearman

Pearson r assumes a straight-line link; **Spearman ρ** uses ranks and does not need normal data. We report both; both were 0.47, so the result does not depend on the method.

9. Partial correlation

The same correlation after "removing" other variables (age, gender, background, chronic disease, trust, AI use). If it stays the same, those variables are not the explanation.

```
r = 0.47 → partial r = 0.48: unchanged
```

> **Say:** *The link stayed the same after adjusting for everything else, so it is not explained by age, gender or background.*

10. Comparing groups: Mann–Whitney and Kruskal–Wallis

To compare a **score** between groups without assuming normality: **Mann–Whitney U** for 2 groups, **Kruskal–Wallis H** for 3 or more.

```
Chronic condition: SHAI 14.2 vs 11.9 → Mann–Whitney p = 0.004
```

11. Chi-square (χ^2)

To test if two **categorical** variables are related, e.g. "elevated health anxiety yes/no" and "uses social media yes/no". It compares the counts we saw with the counts we would expect if there were no link.

12. Many tests → FDR correction (q-value)

If you run 162 tests at p < 0.05, about 8 will be "significant" by pure luck (162 × 0.05 ≈ 8). The **Benjamini–Hochberg FDR** correction adjusts p-values into **q-values**. q < 0.05 = **robust**; p < 0.05 but q ≥ 0.05 = **exploratory** (could be luck).

```
Never disclosing: CSS q = 0.012 (robust) · SHAI q = 0.098 (exploratory)
```

> **Say:** *Because we ran many tests, we corrected for multiple testing; results that did not survive are called exploratory.*

13. Linear regression, B and β

Regression predicts a score (CSS) from several factors at the same time, so each factor is "adjusted" for the others. **B** = how many CSS points change per 1 unit of the factor. **β** = the same on a common scale, so you can compare factors (bigger β = stronger).

```
SHAI: B = 0.68, β = 0.48. Someone with SHAI 20 vs 10 → about 6.8 CSS points
higher.
```

> **Say:** *Each extra point of health anxiety adds about 0.7 points of cyberchondria; it is by far the strongest factor.*

14. R^2, ΔR^2 and hierarchical regression

R^2 = share of differences the model explains (0–1). In **hierarchical** regression we add groups of variables in steps and see how much each step adds (**ΔR^2**).

```
Step 1 demographics R² 0.030 (3%) → step 2 digital use 0.133 (13%) → step 3
health anxiety 0.317 (32%). Health anxiety adds 18.4%.
```

> **Why this was chosen:** Hierarchical order shows that health anxiety matters even after everything else is already in the model.

15. Robust standard errors (HC3)

Regression assumes the "errors" are equally spread. Ours were not (Breusch–Pagan p = 0.004). HC3 robust standard errors fix the p-values and CIs so they are still valid. You only need: "we used robust standard errors because the residuals were uneven".

16. Odds and odds ratios (logistic regression)

Logistic regression predicts a yes/no outcome (elevated health anxiety yes/no). It gives **odds ratios (OR)**: OR 1 = no effect, above 1 = higher odds, below 1 = lower odds. **Adjusted** OR (AOR) = after accounting for the other factors.

```
Worked example from our data:
With elevated health anxiety: 52.6% had high cyberchondria → odds = 52.6/47.4 =
1.11
Without: 23.0% → odds = 23.0/77.0 = 0.30
Odds ratio = 1.11 / 0.30 ≈ 3.7
```

In the model: AOR **3.12** per **1 SD** of cyberchondria (1 SD ≈ 8.4 CSS points) means about three times the odds for every 8 extra points.

> **Trap:** AOR below 1 means protective. Age AOR 0.61: each older age group had lower odds.

17. AUC

How well the logistic model separates people with and without elevated health anxiety. Pick one person with and one without: AUC is the chance the model gives the higher risk to the right one. 0.5 = coin toss, 1 = perfect.

```
AUC 0.83, and 0.79 after correcting for optimism (bootstrap): good
```

> **Why this was chosen:** A model always looks better on the data it was built on, so we corrected the AUC with 500 bootstrap repeats.

18. Bootstrap

A computer trick: draw 5000 new samples of 300 people from our own data (with replacement), redo the analysis each time, and see how much the result varies. That gives a CI without assuming normality. We used it for mediation and to check the regression coefficient.

19. Mediation (indirect effect)

Does part of the link from X to Y go **through** a middle variable M? Path **a** = X → M; path **b** = M → Y (adjusted for X); **indirect** = a × b; **direct** (c′) = what is left.

```
Model A: health anxiety → excessive searching → distress
a = 0.46, b = 0.35 → indirect = 0.46 × 0.35 ≈ 0.16
total = 0.46, direct = 0.30 → about 35% goes through excessive searching
Model C: health anxiety → distress → reassurance seeking: indirect 0.22, direct
0.01 → almost all goes through distress
```

> **Trap:** Mediation in one survey is NOT proof of cause. The reverse order was also significant (0.15). Always say: compatible with a cycle, not proof.

20. Moderation (interaction)

Does a third variable change the **strength** of the link? E.g. is the SHAI–CSS link stronger in AI users? We tested 7 moderators; none were significant after FDR. So the link is similar in everyone.

21. Reliability: Cronbach α and McDonald ω

Do the items of a scale measure the same thing? 0.7 acceptable, 0.8 good, 0.9 excellent.

```
SHAI α 0.81, CSS α 0.86: good
```

22. CFA and fit indices

Confirmatory factor analysis checks whether the items group into the factors the scale claims (CSS-12 = four facets). Fit indices: **CFI** near 0.95 or higher = good; **RMSEA** and **SRMR** 0.08 or lower = acceptable.

```
CSS-12 four factors: CFI 0.95, RMSEA 0.070, SRMR 0.053 → acceptable · SHAI one
factor: CFI 0.83 → poor (it has two factors)
```

23. Measurement invariance

Before comparing Arabic and English answers, we must check the questionnaire works the same in both, like checking two rulers measure the same centimetre. Three steps: **configural** (same shape), **metric** (items equally strong), **scalar** (same starting points). If fit does not get worse at a step (CFI drop ≤ 0.01), that step passes.

CSS-12 passed all three. SHAI-14 passed the first two.

24. DIF

Differential item functioning: does a **single question** behave differently in Arabic vs English for people with the same total score? None did after FDR correction.

25. Sensitivity analysis

Re-doing the main analysis in another way (excluding doubtful responses, removing one item, another method) to see if the result changes. Ours did not change: r stayed 0.46–0.48.

5 · Every table and figure: why it is there and how to read it

The rules that decided every choice

If the data are...	...we used	Example
Exact numbers that people need to look up	a **table**	Table 1 participants, Table 9 regression
A pattern people should see at a glance	a **figure**	Figure 5 scatter plot
Percentages of groups (categorical)	**bar chart**	Figures 3, 4
Two continuous scores together	**scatter plot**	Figure 5
Many correlations at once	**heatmap** (colour = strength)	Figures 6, 11
A score compared across groups, skewed data	**box plot** (median, spread, outliers)	Figure 7
Means of several facets in two groups	**bar chart with 95% CI**	Figure 10
Many odds ratios with CIs	**forest plot**	Figure 8
Paths between variables	**path diagram**	Figure 9

Also: the most important results get **both** a table (exact numbers) and a figure (the picture), e.g. the main correlation (Table 6 + Figure 5) and the logistic model (Table 10 + Figure 8). Less central results have only a table.

Table 1 · Who took part

> **Why this was chosen:** Every study must describe its sample first, so the reader knows who the results apply to. A table, because the reader needs exact n and %.

How to read it:

1. Rows = characteristics (age, gender, education, study/work, background, chronic disease, language).
2. Columns = n (number of people) and %.

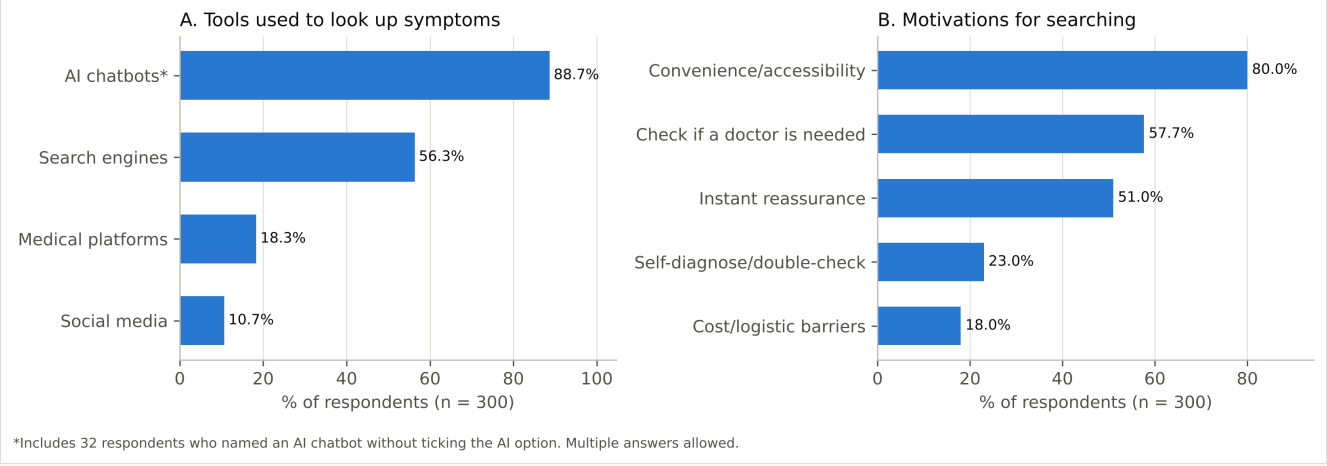
What it shows: A young sample (62.3% aged 20–29), 44.0% medical students, 63.3% with a health background.

> **Say:** *Our 300 participants were mostly young and educated; almost half were medical students. This matters when we interpret percentages.*

> **Trap:** Why so many medical students? Answer: the link spread through our own networks first; we say percentages cannot be generalised.

Table 2 + Figure 3 · How people search and why

> **Why this was chosen:** Tools and motives are categories with multiple answers, so percentages are best shown as bars (Figure 3); Table 2 keeps every exact count.



How to read it:

1. Figure 3, panel A = tools; panel B = motives.
2. Each bar = % of all 300 who ticked that option.
3. Bars add up to more than 100% because people could tick several.

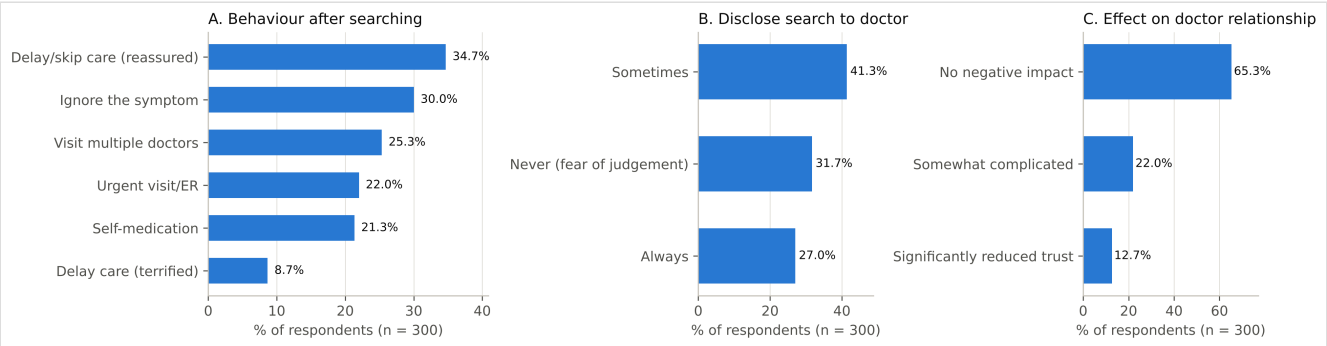
What it shows: AI chatbots 78.0% ticked (88.7% including those who named one) vs search engines 56.3%; convenience is the top motive.

> **Say:** *AI chatbots are now the main gateway: 78.0% ticked them, compared with 56.3% for search engines.*

> **Trap:** Why two AI numbers? 32 people named a chatbot but did not tick "AI"; we report both and test both.

Figure 4 · What people do after searching

> **Why this was chosen:** Three related sets of categories (behaviour, disclosure, relationship) shown side by side so the committee sees the whole consultation story in one picture.



How to read it:

1. Panel A: behaviours after searching (multiple answers).
2. Panel B: do they tell the doctor? (always / sometimes / never).
3. Panel C: effect on the relationship with doctors.

What it shows: 34.7% delay care when reassured; 31.7% never tell the doctor; 34.7% say trust was strained.

> **Say:** *Searching changes the consultation: one in three delays care because they feel reassured, and almost one in three never tells the doctor.*

Table 3 · Scores and reliability

> **Why this was chosen:** Before using any score, we must show the scales work (reliability) and give their levels. A table because these are exact values.

How to read it:

1. Rows: SHAI, CSS and the four CSS facets.
2. Columns: mean ± SD, median (IQR), range, α (95% CI), ω.
3. The note under the table has the CFA fit.

What it shows: Both scales reliable (α 0.81 and 0.86); excessiveness is the highest facet, compulsion the lowest.

> **Say:** *Both questionnaires were reliable. People search a lot, but it rarely disrupts their daily life.*

Table 4 · Prevalence at different cut-offs

> **Why this was chosen:** The % with "elevated" health anxiety depends on the cut-off you choose. Showing four cut-offs is honest and stops the committee from accusing us of choosing the one that looks best.

How to read it:

1. Each row = a cut-off (15, 18, 22, 29).
2. Columns: n, %, 95% CI.
3. Bold row = 18, the one we use.

What it shows: 32.7% at 15, 19.0% at 18, 7.3% at 22, 0.7% at 29.

> **Say:** *About one in five scored 18 or more; this is a screening threshold, not a diagnosis, and the table shows how the estimate changes with the cut-off.*

> **Trap:** Why 18? It best separated people with health anxiety from healthy controls in the Swedish study.

Table 5 · Arabic vs English (measurement invariance)

> **Why this was chosen:** We combined Arabic and English answers. To do that fairly, we had to prove the questionnaire works the same in both languages.

How to read it:

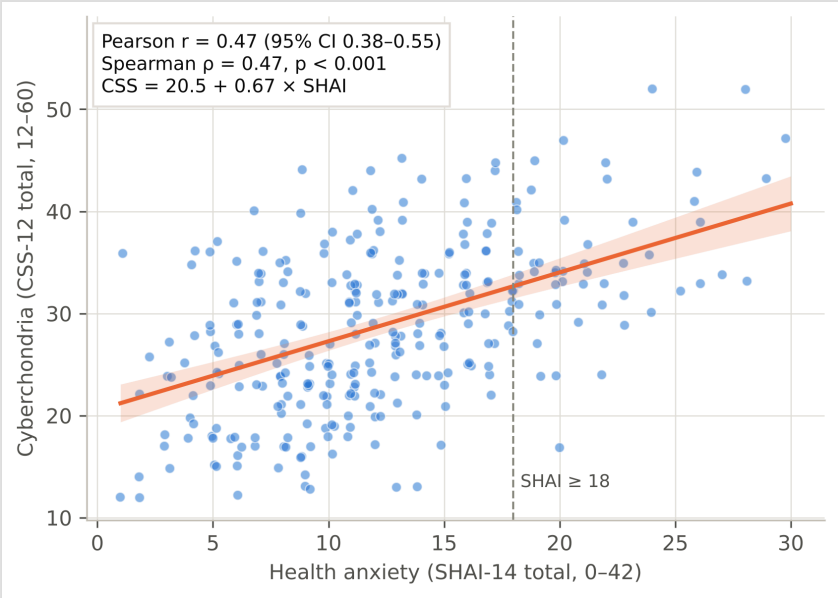
1. Two blocks: CSS-12 and SHAI-14.
2. Three rows each: configural, metric, scalar.
3. Look at ΔCFI and the Decision column: "Supported" = that step passed.

What it shows: CSS-12 passed all steps; SHAI-14 passed metric, not scalar; no item behaved differently.

> **Say:** *The cyberchondria questionnaire works identically in Arabic and English, so combining the two languages was justified.*

Table 6 + Figure 5 · The main result

> **Why this was chosen:** This answers our main question (H1), so it gets both: a figure to see the pattern, a table for exact numbers. A scatter plot is the standard picture for two continuous scores.



How to read it:

1. Each dot = one participant (x = SHAI, y = CSS).
2. The red line = best-fit line; the shaded band = its 95% CI.
3. The dotted vertical line = SHAI 18 cut-off.
4. Table 6: r for the total and for each facet, plus adjusted β.

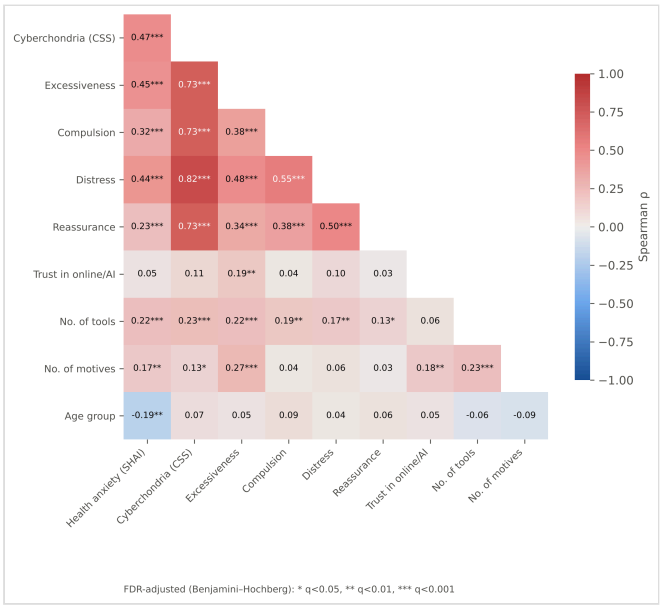
What it shows: Clear upward trend: r = 0.47; strongest for excessiveness (0.46) and distress (0.45).

> **Say:** *This is our main result: the more health anxiety, the more cyberchondria, r = 0.47, and it holds after adjustment.*

> **Trap:** Point at the line going up, then at the r value in the box. Do not say "causes".

Figure 6 · Correlation heatmap

> **Why this was chosen:** We had 10 variables. Writing 45 correlations in text is unreadable; a heatmap shows them all at once, with colour for strength.



How to read it:

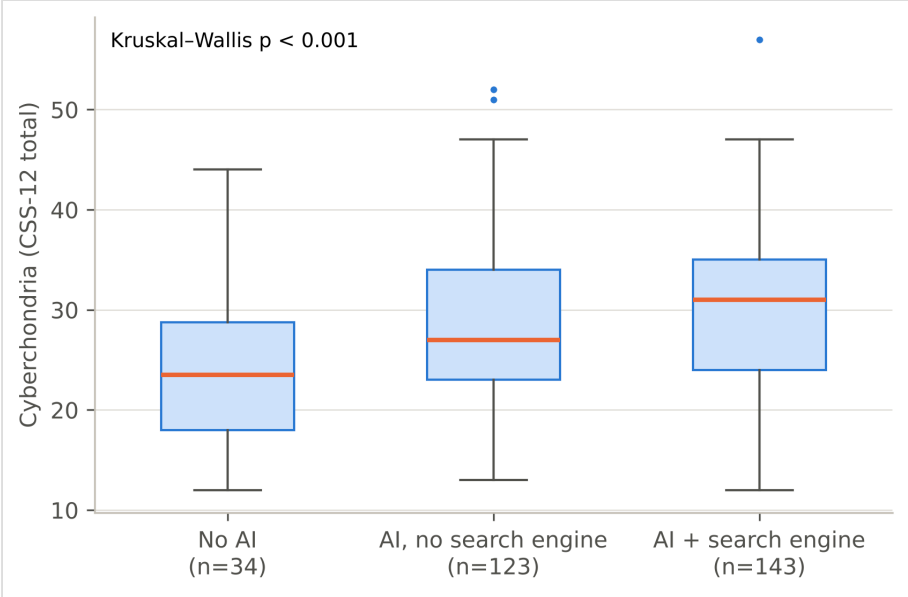
1. Each square = correlation between the row and column variable.
2. Darker red = stronger positive; blue = negative.
3. Stars = significant after FDR (* q<0.05, ** q<0.01, *** q<0.001).

What it shows: Health anxiety correlates with every cyberchondria facet; trust relates only to excessiveness; age is slightly negative.

> **Say:** *The heatmap shows all relationships at once: health anxiety is linked with every part of cyberchondria.*

Table 7 + Figure 7 · Which groups score higher

> **Why this was chosen:** Table 7 compares the scores across every participant group (exact means and p/q). Figure 7 is a box plot because CSS is skewed: it shows the median and spread, not just the mean.



How to read it:

1. Box plot: middle line = median; box = middle half of people; whiskers = range; dots = outliers.
2. Three groups: no AI, AI only, AI + search engine.

What it shows: Health anxiety higher with chronic illness (14.2 vs 11.9) and in reassurance seekers (13.8 vs 10.9); CSS higher in AI users (29.5 vs 24.0), highest in AI + search engine users.

> **Say:** *People who combine AI with search engines had the highest cyberchondria, but we will see this is not robust once we adjust.*

> **Trap:** Do not over-sell the AI difference; it disappears in the strict analysis.

Table 8 · Yes/no outcomes (χ^2)

> **Why this was chosen:** Some outcomes are yes/no (elevated health anxiety, high cyberchondria). χ^2 tests show which categories go with them. Only significant ones are listed to keep it short.

How to read it:

1. Each row: variable, % with the outcome in each group, Cramér's V (strength), p, q.

What it shows: With elevated health anxiety, 52.6% had high cyberchondria vs 23.0% without (OR ≈ 3.7).

> **Say:** *People with elevated health anxiety were much more likely to have high cyberchondria.*

Table 9 · What predicts cyberchondria (regression)

> **Why this was chosen:** Group comparisons look at one factor at a time; regression tests all factors together, so we know which matter independently (H2).

How to read it:

1. Rows = predictors; B = CSS points per unit; β = comparable strength; p.
2. Note under the table = R^2 for each step.

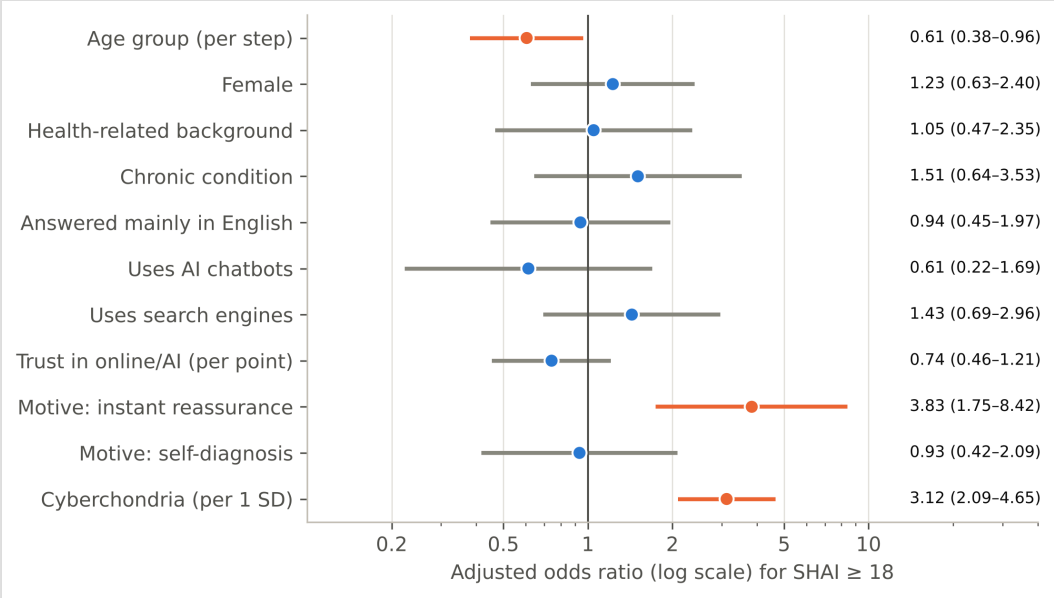
What it shows: Health anxiety β 0.48 (largest by far); R^2 rises from 0.030 to 0.317.

> **Say:** *Demographics explain only 3%; adding health anxiety explains a further 18.4%. Health anxiety is by far the strongest predictor.*

> **Trap:** AI row: significant here, but not with the strict definition (slide 23).

Table 10 + Figure 8 · Who has elevated health anxiety

> **Why this was chosen:** For a yes/no outcome we use logistic regression; a forest plot is the standard way to show many odds ratios with CIs at once.



How to read it:

1. Each dot = AOR; the line = its 95% CI.
2. Vertical line at 1 = no effect. If the line crosses 1 → not significant.
3. Log scale: 0.5 and 2 are the same distance from 1.
4. Orange = significant.

What it shows: Cyberchondria AOR 3.12, reassurance motive 3.83, age 0.61 (protective); AUC 0.83 → 0.79.

> **Say:** *Three factors stand out: higher cyberchondria, searching for reassurance, and younger age.*

Trap: Only 57 events, so the reassurance odds ratio has a wide CI; say “exploratory”.

Table 11 · Behaviour models

Why this was chosen: To test H4 we need adjusted models for each behaviour. A table because there are 8 outcomes × 2 scores: too many for a chart.

How to read it:

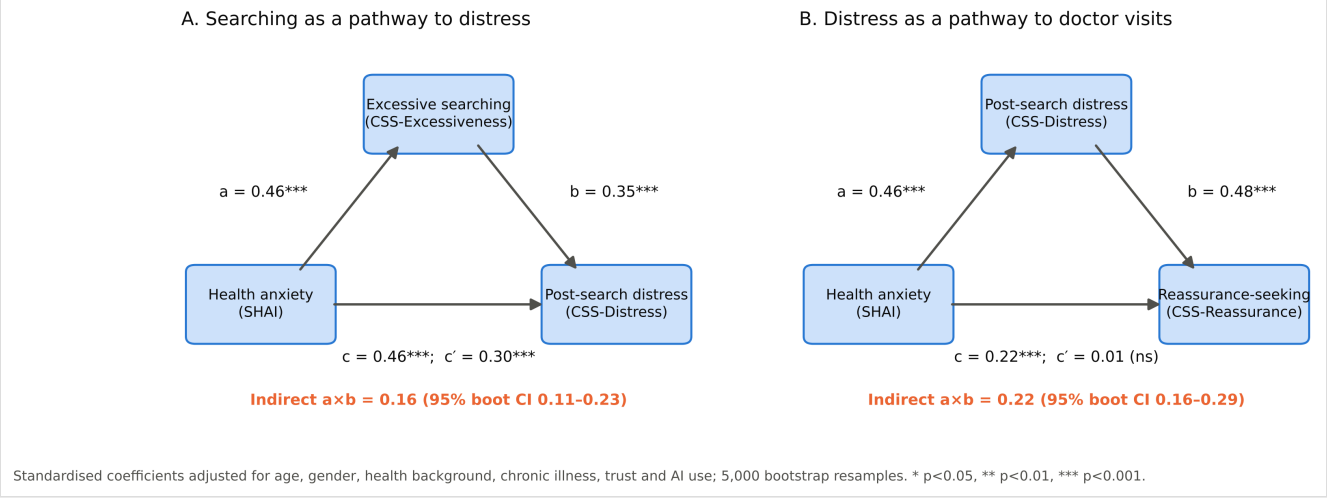
1. Each row = a behaviour; columns = AOR (95% CI) and p for SHAI and for CSS.

What it shows: Higher CSS → urgent visits (AOR 1.51), less ignoring (0.65), less never-disclosing (0.57, the only robust one).

Say: *Cyberchondria, not health anxiety, is what goes with behaviour; only the link with disclosure survived correction.*

Table 12 + Figure 9 · Pathways (mediation)

Why this was chosen: To test H3 (the cycle). A path diagram is the clearest way to show a, b and direct paths; the table adds the CIs and the reverse model.



Standardised coefficients adjusted for age, gender, health background, chronic illness, trust and AI use; 5,000 bootstrap resamples. * p<0.05, ** p<0.01, *** p<0.001.

How to read it:

1. Boxes = variables; arrows = paths with coefficients.
2. a = X → M, b = M → Y, c′ = direct.
3. Below each diagram: the indirect effect with its CI.

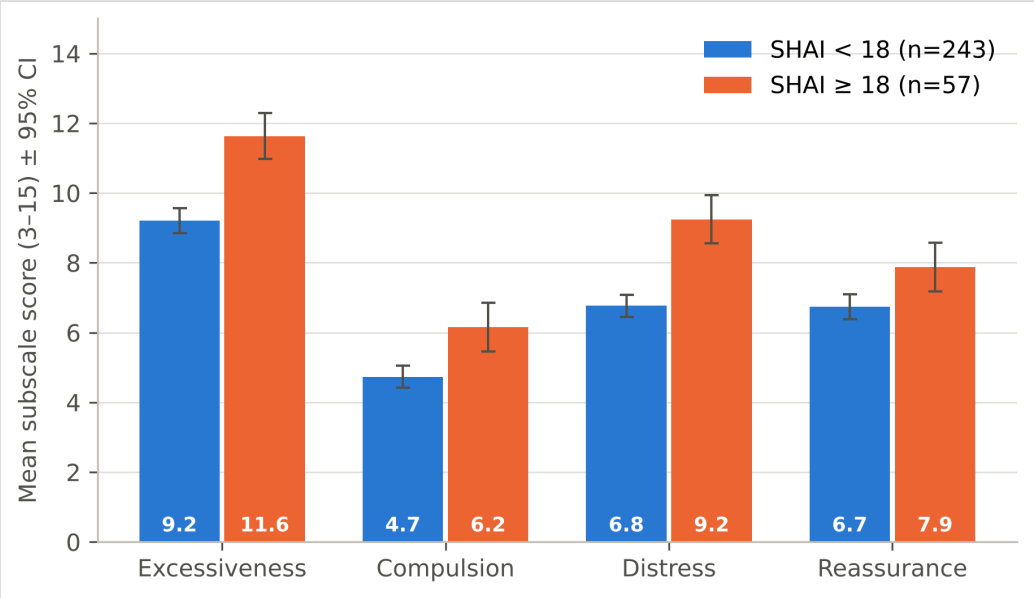
What it shows: Via excessive searching 0.16; via distress 0.22 (direct falls to 0.01); reverse also 0.15.

Say: *The pattern fits a cycle, but because the reverse direction also fits, we cannot say which comes first.*

Trap: Never say “proves”.

Figure 10 · Facets by health-anxiety group

Why this was chosen: Shows simply that people with elevated health anxiety score higher on every facet, and which facets differ most. Bars with 95% CI are best for comparing means of two groups.



How to read it:

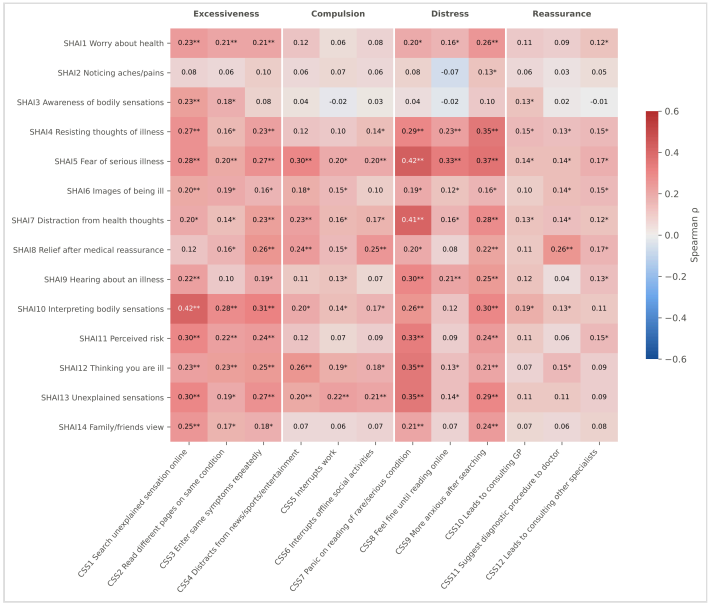
1. Blue = SHAI < 18 (243 people); orange = SHAI ≥ 18 (57 people).
2. Bar height = mean; small lines = 95% CI.

What it shows: Biggest gaps: excessiveness (9.2 vs 11.6) and distress (6.8 vs 9.2).

> **Say:** *People with elevated health anxiety score higher on every part of cyberchondria, most of all on excessive searching and distress.*

Figure 11 (Appendix) · Item by item

> **Why this was chosen:** To see which exact questions drive the link (168 pairs). Only a heatmap can show 168 numbers readably, so it goes in the appendix.



How to read it:

1. Rows = 14 SHAI items; columns = 12 CSS items.
2. Colour = strength; stars = significant after FDR (**: also after the stricter Holm correction).

What it shows: 111/168 pairs significant; strongest: fear of serious illness ↔ panic on reading of a rare condition (0.43). Noticing aches is almost unrelated.

> **Say:** *What drives searching is fear and interpretation, not simply noticing symptoms.*

Table 13 · Sensitivity analyses

> **Why this was chosen:** Committees always ask “is it robust?”. One table answers all such questions at once.

How to read it:

1. Each row = a check; Result; Conclusion column.

What it shows: r stays 0.46–0.48 whatever we exclude; only the AI effect depended on a decision.

> **Say:** *Every check left the main result unchanged.*

6 · Your chapter, section by section

For each section: its **purpose**, the **key points and numbers** (with the reference numbers used in the thesis), and what to **watch**.

3.1 Participants (Table 1)

Who answered.

- 300 adults, no missing data; 62.3% aged 20–29; 56.3% female; most current students; medical students 44.0%; health background 63.3%; chronic condition 21.7% (migraine, hypertension, asthma…); Arabic 70.7%.

Watch: Message: a young, educated, digitally active sample.

3.2 Online health-information seeking (Table 2, Figure 3)

How people search.

- AI ticked 78.0% (88.7% incl. named) vs search engines 56.3%; AI without any search engine 41.0%; ChatGPT 59.3%.
- AI use 96% at 18–19 y, 72% at ≥ 40 y; the 34 non-users are older.
- Motives: convenience 80.0%, "do I need a doctor?" 57.7%, reassurance 51.0%; trust mostly moderate.

Watch: Message: for this sample, the first "doctor" is often an AI.

3.3 Behaviour and the doctor relationship (Figure 4)

- Delay/skip care when reassured 34.7%; ignore 30.0%; several doctors 25.3%; urgent care 22.0%.
- Always tell the doctor 27.0%; never (fear of judgement) 31.7%; relationship complicated or trust reduced 34.7%.

Watch: The "never" option combined frequency and a reason: say "never tell, citing fear of judgement".

3.4 Levels and psychometrics (Tables 3, 4, 5)

- SHAI 12.3 ± 5.8; elevated 19% (CI 15.0–23.8%); Table 4: 32.7% at ≥ 15, 19.0% at ≥ 18, 7.3% at ≥ 22, 0.7% at ≥ 29.
- CSS 28.9 ± 8.4; excessiveness 9.66 (highest), compulsion 5.00 (lowest).
- Reliability α 0.81 / 0.86; ω 0.82 / 0.87. CSS-12 four-factor CFA (ML) CFI 0.95, RMSEA 0.070, SRMR 0.053; SHAI one-factor poor (CFI 0.83): not strictly unidimensional.
- Table 5: CSS-12 configural → metric ($\Delta\chi^2$ 5.52, p 0.70, ΔCFI +0.002) → scalar ($\Delta\chi^2$ 8.41, p 0.40, ΔCFI −0.001): fully invariant. Both baselines are imperfect (configural CFI 0.937 CSS, 0.808 SHAI), so the SHAI result is hard to interpret. SHAI-14 metric supported (p 0.73), scalar not (p 0.007, ΔCFI −0.018). No item DIF after FDR.

Watch: Explain why prevalence depends on the cut-off and why that is not a weakness: it is honest reporting.

3.5 Main association (Table 6, Figures 5, 6)

The central result.

- r = 0.47 (95% CI 0.38–0.55), ρ = 0.47, explains 22%; partial r = 0.48.
- Facets: excessiveness 0.46, distress 0.45 (stronger, Williams' t p < 0.001) vs compulsion 0.25, reassurance 0.24.
- Item level: fear of serious illness and interpretation of sensations matter, mere awareness does not; 111/168 pairs survive FDR (58 Holm); strongest ρ 0.43 (Figure 11, Appendix A).

Watch: H1 and H2 are answered here.

3.6 Bivariate factors (Tables 7, 8, Figure 7)

162 tests.

- Health anxiety higher with chronic condition (p 0.004, q 0.028) and reassurance motive; age/education/occupation borderline (exploratory); gender no difference.

- AI users higher CSS (inclusive), mainly excessiveness and reassurance; highest in AI + search engine users; chatbot brand no difference.
- χ^2: elevated HA → 52.6% vs 23.0% high cyberchondria (OR ≈ 3.7).

Watch: Remember q-values: “survives correction” vs “exploratory”.

3.7 Multivariable models (Tables 9, 10, Figure 8)

- Hierarchical regression: R^2 0.030 → 0.133 → 0.317; health anxiety adds 18.4% (f^2 medium); B = 0.68, β = 0.48.
- AI: +5.3 (CI 2.3–8.3) inclusive vs +1.5, p 0.31 strict; the 32 reclassified had the highest CSS (31.6 vs 24.0 in non-users).
- Reverse model: CSS strongest predictor of SHAI; chronic disease B +1.7 (p 0.029); medical students no longer differ after adjustment.
- Logistic (57 events): AOR 3.12 cyberchondria per SD, 3.83 reassurance motive, 0.61 age; AUC 0.83 → 0.79; ≈ 5 events per variable → exploratory.

Watch: You will be asked to explain AOR, AUC and why it was corrected.

3.8 Behaviour models (Table 11)

- Cyberchondria (not health anxiety) goes with urgent visits (AOR 1.51), less ignoring (0.65), strained relationship (ordinal OR 1.34).
- Only robust behavioural result: higher CSS → lower odds of never disclosing (AOR 0.57, q 0.012). Health anxiety → concealment AOR 1.47, q 0.098 (exploratory).
- Trust → fewer multiple-doctor visits (AOR 0.59).

Watch: H4 only partially supported.

3.9 Pathways (Table 12, Figures 9, 10)

- HA → excessive searching → distress: indirect 0.16 (CI 0.11–0.23), direct 0.30.
- HA → distress → reassurance seeking: indirect 0.22 (CI 0.16–0.29), direct 0.01 (ns).
- Reverse path also significant (0.15); serial path 0.07. No moderator significant after FDR; age trend p 0.07.

Watch: Always end with: “compatible with a cycle, not proof of direction”.

3.10 Sensitivity (Table 13)

- Excluding annotated (r 0.46) or straight-liners (r 0.47), dropping SHAI item 2 (r 0.48), bootstrap CI for B 0.54–0.83, quantile B 0.55–0.85, no quadratic effect (p 0.25), VIF ≤ 3.9, early vs late responders no difference; residuals non-normal → robust SEs.
- Only the AI effect depended on a decision (definition).

7 · Statistical analysis (Section 2.6, word for word)

You are the team's statistics person. Read this until you can explain each method in one sentence (Section 4 gives the plain-language version of each).

Analyses were carried out in Python 3 (pandas, SciPy, statsmodels, pingouin, factor_analyzer and semopy). The analysis code is fully scripted and reproducible. All tests were two-sided with $\alpha = 0.05$.

Descriptive statistics: frequencies and percentages for categorical variables; mean ± SD and median (IQR) for scores. Normality was assessed with the Shapiro–Wilk test, skewness and kurtosis.

Psychometrics: internal consistency by Cronbach's α (with 95% CI) and McDonald's ω [47]; corrected item–rest correlations; sampling adequacy (KMO, Bartlett's test); and confirmatory factor analysis by maximum likelihood (ML), with fit judged by CFI, TLI, RMSEA and SRMR against conventional criteria (CFI/TLI ≥ 0.95, RMSEA ≤ 0.06–0.08, SRMR ≤ 0.08), recognising that fixed cut-offs do not transfer exactly to every model [48]. Diagonally weighted least squares (DWLS) estimates on Pearson covariances are reported for comparison only, because their fit indices are not comparable with these cut-offs. Equivalence of the Arabic and English responders was tested by comparing α values with Feldt's test, and adjusted mean differences. Formal measurement invariance was then tested by multi-group CFA (ML with a mean structure), fitting configural, metric (equal loadings) and scalar (equal loadings and intercepts) models in sequence; each step was judged by the χ^2 difference test and by a change in CFI of no more than −0.01 and in RMSEA of no more than 0.015 [49]. Differential item functioning (DIF) was examined for every item by ordinal logistic regression of the item on the scale total, language and their interaction, a regression-based approach that performs well for ordinal items [50], with FDR correction across the 26 items.

Correlation analysis: Pearson and Spearman correlations (bootstrapped 95% CIs) and partial correlations adjusted for covariates. Differences between dependent correlations (health anxiety with each CSS subscale) were tested with Williams' T2 test for dependent correlations, as recommended by Steiger [51]. Item-level correlations were computed for every SHAI item with the CSS total and every CSS item with the SHAI total.

Group comparisons for every variable: Mann–Whitney U (rank-biserial r as effect size) or Kruskal–Wallis H (ε^2 as effect size) for SHAI, CSS and each CSS subscale, with Welch's t-test/ANOVA as parametric confirmation and pairwise Mann–Whitney post-hoc comparisons with Holm adjustment. Associations of all categorical variables with elevated health anxiety and high cyberchondria were tested with Pearson χ^2 or Fisher's exact test (Monte-Carlo exact p when expected counts were small), with Cramér's V. Pairwise associations among all categorical variables were also screened.

Multiple testing: because many associations were tested, p-values within each family of tests were adjusted with the Benjamini–Hochberg false discovery rate procedure (q-values) [52]. The families were: the 162 group comparisons; the chi-square tests; the pairwise associations among categorical variables; the 40 coefficients of the behaviour models; the moderation terms; and the 168 item-level correlations. Unadjusted p-values are reported, and associations that do not survive correction are labelled as exploratory.

Multivariable models: hierarchical OLS regression of CSS on socio-demographics (block 1), digital-use variables (block 2) and SHAI (block 3), with HC3 robust standard errors, standardised coefficients and VIF checks; the reverse model (SHAI on CSS); models for each CSS subscale; quantile regression (25th–90th percentiles) and a quadratic term to examine non-linearity. Logistic regression identified predictors of elevated health anxiety (AORs; apparent AUC and AUC corrected for optimism by 500 bootstrap refits) and of each pre-consultation behaviour; ordinal logistic regression was used for the ordered disclosure and relationship outcomes.

Mediation and moderation: as an exploratory decomposition of the association into CSS facets, mediation models with bootstrap confidence intervals (5000 resamples) estimated how much of the link between health anxiety and post-search distress ran through excessive searching, and related pathways, including a reverse-direction model and

a serial model [53]. Moderation of the SHAI–CSS association by trust, AI use, health background, gender, chronic disease, language and age was tested with interaction terms and simple slopes.

Sensitivity analyses: exclusion of annotated responses; exclusion of straight-line responders (identical answers to all items of a scale); alternative SHAI cut-offs; bootstrap of the main regression coefficient; and comparison of early versus late responders as a proxy for non-response bias.

The flow of the study, from questionnaire development to the stages of analysis, is summarised in Figure 2.

8 · Your slides

A. What to point at and what to say (short version)

Short versions (three sentences each). The full scripts are in the speaker notes and in your Student 2 guide. Aim for 30–35 seconds per slide.

Slide	Point at	Say (short version)
13 · Results divider	Nothing to point at.	Thank you. I will present our results in six steps: who took part, how they search, the levels of health anxiety and cyberchondria, the main association, the predictors and pathways, and behaviour and AI.
14 · Participants (Table 1)	The highlighted rows: gender and medical students.	Table 1 shows our 300 participants. Most were young, 62.3% aged 20 to 29, and 44.0% were medical students. In summary, a young and educated sample, which we keep in mind when reading percentages.
15 · How people search (Figure 3)	The top bar of panel A (AI chatbots), then the four number boxes.	Figure 3 shows the tools people use. AI chatbots come first, 78.0%, ahead of search engines at 56.3%, and ChatGPT alone is used by 59.3%. In summary, for many people the first “doctor” is now an AI chatbot.
16 · Levels & reliability (Table 3)	The first two rows, then the 19% box.	Table 3 shows the scores. 19% scored in the elevated range of health anxiety, and both questionnaires were reliable, with alpha 0.81 and 0.86. In summary, our tools measured well, in both Arabic and English.
17 · Main result (Figure 5)	The rising line, then the r box.	This is our main result. Health anxiety and cyberchondria rise together, r = 0.47, and the link stays 0.48 after adjusting for everything else. In summary, hypotheses 1 and 2 are supported.
18 · What predicts cyberchondria (Table 9)	The R^2 rows, then the 18.4% box.	Table 9 is our regression. Demographics explain only 3% of cyberchondria, but health anxiety adds another 18.4%. In summary, how anxious you are matters far more than who you are.
19 · Item by item (Figure 11)	The 0.43 and 0.42 boxes, then the “≈ 0” box.	Figure 11 looks at every pair of questions. The strongest links are fear of serious illness and interpreting body sensations; simply noticing aches is almost unrelated. In summary, fear drives searching, not noticing.
20 · Who has elevated health anxiety (Figure 8)	The orange dots, then the four boxes.	Figure 8 shows the odds ratios. Higher cyberchondria triples the odds, searching for reassurance nearly quadruples them, and older age is protective; the model's AUC is 0.79 after correction. In summary, the person most at risk is young and searches for reassurance.
21 · Pathways (Figure 9)	Panel A indirect effect, then panel B direct effect, then the orange box.	Figure 9 shows the pathways. Part of the link runs through excessive searching (0.16), and almost all of the link to reassurance seeking runs through distress. In summary, this is compatible with a cycle, but one survey cannot prove the direction.
22 · Behaviour (Figure 4)	The first bar of panel A, then the four boxes.	Figure 4 shows what people do after searching: 34.7% delay care because they feel reassured, and 31.7% never tell their doctor. Higher cyberchondria goes with more disclosure, and that result survived correction. In summary, searching changes the consultation before it begins.
23 · AI chatbots	The table: inclusive vs strict rows.	Finally, AI. AI users scored 5.3 points higher, but with the strict definition the difference was only 1.5 and not significant, p = 0.31. In summary, there is no reliable independent effect of AI. I now hand over to my colleague.

B. Full scripts (speaker notes)

The script below is the speaker-notes text of your slides in the presentation. Practise until you can say it in your own words without reading.

Slide 13 · Results

Thank you. I will present the results in six steps: who took part, how they search, the levels of health anxiety and cyberchondria, the main association, the predictors and pathways, and finally behaviour and AI chatbots.

Slide 14 · Participants

Opening: Table 1 shows our 300 participants. Main finding: the sample is young, 62.3% aged 20 to 29, slightly more women at 56.3%, and 44.0% are medical students; 63.3% have a health-related background, and 21.7% reported a chronic condition. 70.7% answered mainly in Arabic. Closing: so this is a young, educated, digitally active sample, and that matters when we interpret percentages.

Slide 15 · How People Search

Opening: Figure 3 shows the tools people use and why. Main finding: AI chatbots are now the main gateway. 78.0% ticked AI chatbots, 88.7% if we include those who named one, compared with 56.3% for search engines. 41.0% use AI with no search engine at all, and ChatGPT alone is used by 59.3%. The top motives are convenience at 80.0%, checking whether a doctor is needed at 57.7%, and instant reassurance at 51.0%. Closing: in summary, for this sample, the first doctor is often an AI chatbot.

Slide 16 · Levels & Reliability

Opening: Table 3 shows the scores and reliability. Main finding: mean SHAI was 12.3 out of 42 and 19% scored 18 or more, our threshold for elevated health anxiety, with a 95% confidence interval of 15.0–23.8%. Mean CSS was 28.9 out of 60. Excessiveness was the highest facet and compulsion the lowest: people search a lot, but it rarely disrupts their lives. Both scales were reliable, alpha 0.81 and 0.86, and reliability was the same in Arabic and English responders. Formal invariance testing showed the CSS-12 works the same way in both languages, and no single item behaved differently. Closing: our tools measured well in this population.

Slide 17 · Health Anxiety ↔ Cyberchondria

Opening: this is our main result, Figure 5. Main finding: health anxiety and cyberchondria rise together, r = 0.47, 95% CI 0.38–0.55, so health anxiety explains about 22% of the variation in cyberchondria. After adjusting for every other variable, the partial correlation is still 0.48. The link is strongest for excessive searching, 0.46, and distress, 0.45, and weaker for compulsion and reassurance. Closing: in summary, hypotheses 1 and 2 are supported.

Slide 18 · What Predicts Cyberchondria?

Opening: Table 9 is the hierarchical regression for cyberchondria. Main finding: demographics alone explain only about 3% of cyberchondria. Adding digital-use variables brings it to 13%. Adding health anxiety adds another 18.4%, to about 32%. Health anxiety is the strongest predictor, standardised beta 0.48: each extra SHAI point adds about 0.68 CSS points. Closing: who you are matters little; how anxious you are matters most.

Slide 19 · Item by Item: What Drives Searching?

Opening: Figure 11 breaks the association down item by item, all 168 pairs. Main finding: 111 pairs remained significant after FDR correction. The strongest links involve fear of serious illness and interpreting body sensations; for example fear of illness with panicking when reading about a rare condition. Simply noticing aches and pains was almost unrelated. Closing: what drives searching is fear and interpretation, not just noticing symptoms, exactly what the cognitive model predicts.

Slide 20 · Who Has Elevated Health Anxiety?

Opening: Figure 8 shows who has elevated health anxiety, in a multivariable logistic model. Main finding: three factors are independent. Each standard deviation of cyberchondria triples the odds, adjusted odds ratio 3.12. Searching for instant reassurance almost quadruples them, 3.83. Older age is protective, 0.61 per age step. The model discriminates well: AUC 0.83, 0.79 after correcting for optimism. Closing: the person most at risk is young and searches to feel reassured.

Slide 21 · Pathways: Is There a Cycle?

Opening: Figure 9 shows the pathways. Main finding: in model A, part of the link from health anxiety to post-search distress runs through excessive searching, indirect effect 0.16, CI 0.11–0.23. In model B, post-search distress carries almost all of the link from health anxiety to reassurance-seeking visits: the direct effect drops to 0.01. But the reverse direction also fits, 0.15. Closing: our data are compatible with the cycle, but one survey at one time point cannot prove direction. Hypothesis 3 is supported statistically, not causally.

Slide 22 · Behaviour Before the Consultation

Opening: Figure 4 shows what people do after searching. Main finding: the most common response is to delay or skip care because they feel reassured, 34.7%. Others escalate: urgent visits 22.0%, multiple doctors 25.3%. Only 27.0% always tell their doctor about their searches, and 31.7% never do, for fear of being judged. Higher cyberchondria was linked with more disclosure, odds ratio 0.57, and this survived FDR correction. 34.7% say searching complicated or reduced trust in their doctor. Closing: searching changes the consultation before it begins.

Slide 23 · AI Chatbots: An Independent Effect?

Opening: finally, did AI chatbots themselves matter? Main finding: AI users scored 5.3 points higher on cyberchondria after adjustment, mainly excessiveness and reassurance. But 32 people named a chatbot without ticking the AI option. When we use the strict definition, only people who ticked it, the effect disappears: B = 1.5, p = 0.31. And only 34 people were true non-users. AI use also did not change the strength of the anxiety–cyberchondria link. Closing: no reliable independent effect of AI. Absence of evidence, not evidence of absence. I now hand over to my colleague for the discussion.

> **Delivery for your part:** ≈ 6 minutes; about 25–35 seconds per content slide and 10 seconds per divider. Point at the number you are saying. Pause after each key number. End with: “Those are our results. My colleague will now discuss what they mean.”

9 · Questions and model answers

A. Questions on your part

Q: Why non-parametric tests?

Shapiro–Wilk showed the scores were not normally distributed; Mann–Whitney and Kruskal–Wallis do not assume normality. Welch tests confirmed the same conclusions.

Q: What is FDR and why did you use it?

With 162 comparisons, about 8 would be "significant" by chance at $p < 0.05$. Benjamini–Hochberg controls the proportion of false discoveries; results with $q \geq 0.05$ are labelled exploratory.

Q: Explain r = 0.47 to a non-statistician.

People with more health anxiety clearly tend to have more cyberchondria; anxiety accounts for about a fifth (22%) of the differences.

Q: Why both Pearson and Spearman?

Pearson for the linear correlation, Spearman as a check that does not depend on normality; both were 0.47.

Q: What does partial r = 0.48 tell us?

The link is not explained by age, gender, background, chronic disease, trust or AI use: it survives adjustment (H2).

Q: Why hierarchical regression?

To see what health anxiety adds after demographics and digital use: 18.4% extra variance, by far the biggest block.

Q: What do B and β mean in Table 9?

B = 0.68: each SHAI point adds about 0.7 CSS points. β = 0.48: the standardised effect, used to compare predictors.

Q: Why robust (HC3) standard errors?

The residuals were heteroskedastic (Breusch–Pagan $p = 0.004$); HC3 gives valid standard errors anyway.

Q: What is an adjusted odds ratio of 3.12?

For each standard deviation higher cyberchondria, the odds of elevated health anxiety are about three times higher, holding other factors constant.

Q: Why is the reassurance-motive AOR so wide?

Only 57 events for 11 predictors (≈ 5 per variable), so estimates are imprecise and probably optimistic; we label them exploratory.

Q: What is the AUC and why corrected?

How well the model separates elevated from non-elevated anxiety. Models look better on their own data, so 500 bootstrap refits corrected 0.83 to 0.79.

Q: Does the mediation prove a vicious cycle?

No. One time point, the reverse order fits too, and the facets come from one questionnaire. It is compatible with a cycle; only longitudinal data can show direction.

Q: Why does the AI effect disappear?

It depends on 32 people who named a chatbot without ticking it; they had the highest scores. Under the strict definition p = 0.31. Only 34 non-users made a small comparison group.

Q: Chronic illness was not significant in the logistic model. Why?

Dichotomising SHAI at 18 loses information and power. On the continuous score chronic illness is significant: p = 0.004 unadjusted and p = 0.029 adjusted (B +1.7).

Q: What do CFI, RMSEA and SRMR mean?

Fit indices for CFA. CSS-12 four factors: CFI 0.95, RMSEA 0.070, SRMR 0.053: acceptable. SHAI one-factor fitted poorly.

Q: Why does the SHAI fit poorly?

It is not strictly unidimensional; recent validations find two factors (illness likelihood and consequences). Reliability was still good (α 0.81).

Q: What is measurement invariance (Table 5)?

Testing in steps whether the scale works the same in Arabic and English: same structure, then same loadings, then same intercepts. CSS-12 passed all three; SHAI-14 passed the first two.

Q: Why is high cyberchondria defined as > 33?

There is no validated clinical cut-off for the CSS-12, so we used the upper tertile of our sample, as is common.

Q: Why is the prevalence so dependent on the cut-off (Table 4)?

Because most scores cluster just below the thresholds; that is why we report several cut-offs and say “elevated”, not “disorder”.

Q: What does Table 13 show?

That the main correlation stays 0.46–0.48 whatever we exclude, and the regression is stable. Only the AI effect depended on a decision.

B. Questions anyone can be asked (know the answer even if it is not your part)

Q: Summarise your study in one minute.

Use the two-minute script in Part 1, shortened: design, sample, two scales, r = 0.47, AI result, behaviour, clinical message.

Q: What is your main finding?

Health anxiety and cyberchondria are moderately and robustly associated (r = 0.47), independent of other factors.

Q: What did you find about AI chatbots?

They are the main source (78.0% ticked), but there was no reliable independent effect on cyberchondria beyond health anxiety.

Q: What percentage had elevated health anxiety?

19% at SHAI ≥ 18, a screening threshold, not a diagnosis.

Q: What is your biggest limitation?

The cross-sectional design: no direction of cause.

Q: What is the clinical message?

Ask patients, without judgement, what they searched or asked an AI.

Q: Why did you split the presentation this way?

By chapter, but each of us studied the whole thesis.

Q: Were your hypotheses written before you collected the data?

Answer truthfully. A safe true answer: they follow from our objectives and previous research, and the questionnaire was designed to measure these things; we tested all of them and report every result, including those not supported.

Watch out: Never claim they were pre-registered.

Q: Who did the statistical analysis?

Answer honestly according to what you did. If AI tools helped with code or drafting, say so in line with your college policy and the declaration in the thesis.

C. Difficult or hostile questions

Q: "Your p-values are tiny because your sample is large."

The effect sizes are meaningful too: r = 0.47 is moderate and ΔR^2 18.4% is a medium effect ($f^2 \approx 0.27$).

Q: "You ran so many tests; something will be significant."

Exactly why we used FDR correction and separated robust from exploratory findings.

Q: "The mediators are subscales of the same scale. Isn't that circular?"

Partly, yes: shared method variance inflates the paths. That is why we call it an exploratory decomposition, not a mechanism.

D. Limitations and what was done about each (all students)

Limitation	What we did / how to answer
Cross-sectional design	Reverse-direction models; wording "compatible with"; longitudinal studies recommended
Convenience/snowball sample	Percentages not generalised; associations consistent across subgroups
Arabic wording (no back-translation)	Equal reliability and means; CSS-12 fully invariant, SHAI-14 metric; no item DIF
Single-item behaviour measures	Reported descriptively; associations labelled exploratory
Many tests	FDR correction; robust vs exploratory separated
Small subgroups (34 non-users)	Acknowledged; AI effect called "not robust"
Self-report and careless responding	Anonymous; straight-liners and annotated responses excluded in sensitivity analyses (no change)

> **How to answer any question:** pause → repeat the question briefly → answer in two or three sentences → stop. If you do not know: "We did not examine that directly; based on our data we would expect…, and it is a good question for future research." Never invent a number. If a teammate struggles: "If I may add…".

E. Drills: explain it simply

Cover the right column. Answer out loud in one or two sentences. Then check.

Explain…	Model answer
What is a p-value?	The chance of getting a result this strong by luck if there were really no effect. Below 0.05 we call it significant.

What does r = 0.47 mean?	When health anxiety goes up, cyberchondria tends to go up too; a moderate link, explaining about a fifth of the differences.
What is a confidence interval?	The range where the true value probably lies. Ours for r is 0.38 to 0.55.
Why did you correct for multiple tests?	With 162 tests, about 8 would be significant just by luck; FDR correction separates real results from probable luck.
What is regression for?	To see which factors matter when we look at all of them together.
What is β?	The strength of each factor on a common scale, so we can compare them. Health anxiety had the biggest, 0.48.
What is an odds ratio of 3?	The odds are three times higher. Below 1 means lower odds.
What is AUC?	How well our model tells anxious from non-anxious people apart: 0.79 is good.
What is mediation?	Checking whether part of a link passes through a middle step, here searching or distress.
Does mediation prove a cycle?	No. The reverse also fits, and it is one survey at one time point.
Why did the AI effect disappear?	It depended on 32 people who named a chatbot but did not tick the box; only 34 non-users. With the strict definition it was not significant.
Why non-parametric tests?	Our scores were not perfectly normal; these tests do not need that.
What is Cronbach's alpha?	Whether the questions of a scale measure the same thing; ours were 0.81 and 0.86, good.
What is measurement invariance?	Checking the questionnaire measures the same thing in Arabic and English, like checking two rulers. The CSS-12 passed fully.
What is a sensitivity analysis?	Doing the analysis in other ways to see if the result changes. It did not.
Why both a table and a figure for some results?	The figure shows the pattern; the table gives the exact numbers. We did both for the most important results.
Why a box plot for Figure 7?	Cyberchondria scores are skewed, so the median and spread are more honest than a bar of means.
Why a forest plot for Figure 8?	It shows many odds ratios and their CIs at once; anything crossing 1 is not significant.
Why a heatmap for Figure 6?	There are 45 correlations; colour lets you see them all at once.
Why is Figure 11 in the appendix?	It has 168 numbers: important detail, but too much for the main chapter.

10 · Your key sources

Every reference cited in your chapter, with its number in the thesis. Know the bold ones well: examiners often ask “who found that?”.

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[53]	Hayes AF. Introduction to mediation, moderation, and conditional process analysis: a regression-based approach. 3rd ed. New York: Guilford Press; 2022.

11 · Editing your chapter

> **The Word file is the master copy.** Edit only the .docx, in one shared OneDrive / Word Online file with **Review → Track Changes** on. Keep an untouched copy (Thesis_ORIGINAL.docx). Avoid Google Docs: it breaks the automatic numbering.

For your chapter

- Results is mostly numbers: **do not change any number**, p, q, CI, test name or table reference.
- You may reword the connecting sentences in your own words; keep one main message per paragraph.
- Never type table or figure numbers by hand in captions; in text, keep "Table n / Figure n" exactly.
- If a number or sentence looks wrong, add a comment and ask; do not "fix" statistics yourself.
- Read 2.6 Statistical analysis until you can explain each method in one sentence; you will be asked.

Never touch

- Numbers, %, p, q, CI, r, OR, β, n
- Reference numbers like [18] (move them with their sentence)
- The reference list
- Table/figure captions, table of contents and lists (automatic fields)
- Heading numbers and styles
- Times New Roman 14 and page numbering

The rewrite method (paragraph by paragraph)

1. Read until you understand it; look up anything unclear.
2. Close it and explain it out loud.
3. Write your version from that explanation.
4. Compare: same facts, numbers and [n]; nothing claimed more strongly.
5. Read aloud; change any phrase you would never say.

After editing

1. Ctrl + A, then F9, to update the table of contents, the lists and the captions.
2. Search for "[" to find placeholders left.
3. Swap chapters with a teammate for checking.
4. Export to PDF and look at every page.

> **About AI use, honestly.** Check your college's AI policy with your supervisor and declare AI assistance if required. A sentence you can adapt: *"The statistical analysis code and an initial draft of this report were prepared with the assistance of an AI tool (Claude, Anthropic). All analyses were checked, and the text was reviewed and rewritten by the authors, who take full responsibility for its content."* Do not use paraphrasing tools or tricks to fool detectors: they make the text worse and can count as misconduct. Real ownership (your words, your understanding, your edit history) is the best protection.

12 · Plan, self-test and checklist

14-day plan

Day	Task
1	Read the Abstract, 1.8 and Chapter 5. Learn the two-minute script (Part 1).
2	Read the whole thesis once, marking what you cannot explain.
3	Core facts and terms (Part 1). Self-test on core facts.
4	Your chapter, sections in Part 2 (first half).
5	Your chapter, second half; your key sources (Part 3).
6	Start rewriting your chapter (Part 6).
7	Finish rewriting. Self-test (Part 8).
8	Swap chapters with a teammate; check accuracy and voice.
9	Your slides: read the scripts (Part 4), then present them alone 3 times.
10	Questions A–D out loud (Part 5). Record yourself once.
11	Team: full rehearsal with timing and handovers.
12	Team: mock committee; each student asks the others their Part 5 questions.
13	Fix weak answers; second full rehearsal; final Word checks.
14	Day before: checklist (last page); one calm run-through; sleep.

Self-test

Cover the answers, answer out loud, then check. Repeat until all are correct.

#	Question	Answer
1	Main correlation with 95% CI?	r = 0.47 (0.38–0.55)
2	Partial correlation?	0.48
3	Which two facets correlate most with health anxiety?	Excessiveness (0.46) and distress (0.45)
4	R^2 of the three regression blocks?	0.030 → 0.133 → 0.317
5	β and B for health anxiety in the final model?	β 0.48, B 0.68
6	Three independent predictors of elevated health anxiety and their AORs?	Cyberchondria 3.12, reassurance motive 3.83, age 0.61
7	AUC apparent and corrected?	0.83 → 0.79
8	Indirect effect via excessive searching?	0.16 (CI 0.11–0.23)
9	Direct effect of HA on reassurance seeking after distress?	0.01 (not significant)
10	The only behavioural association that survived FDR?	Higher cyberchondria → less never-disclosure (AOR 0.57, q 0.012)
11	AI effect inclusive vs strict?	+5.3 vs +1.5 (p 0.31)
12	Prevalence at cut-offs 15/18/22/29?	32.7% / 19.0% / 7.3% / 0.7%
13	Which scale is fully invariant across languages?	CSS-12 (scalar); SHAI-14 metric only
14	How many of 168 item pairs survived FDR?	111

15	Reliability α of both scales?	0.81 and 0.86
16	Main correlation (everyone)?	r = 0.47
17	Elevated health anxiety (everyone)?	19%
18	AI chatbots ticked (everyone)?	78.0%
19	Never tell the doctor (everyone)?	31.7%
20	Sample size (everyone)?	300

Day-before checklist

✓	Item
☐	All placeholders filled (names, supervisor, college, logo, degree, year, ethics, Google Forms question, Acknowledgment)
☐	Fields updated (Ctrl + A, F9); PDF checked page by page
☐	AI declaration added if required
☐	Presentation: names on slide 1, ethics number on slide 10; USB + PDF + email copies; clicker tested
☐	I can give the two-minute summary and my slides (13–23) without notes
☐	I have answered every question in Part 5 out loud
☐	I scored 100% on the self-test
☐	Printed thesis copy if required; arrive early; agree who stands where

Generated from the study outputs and the presentation, so every number and slide script here matches the thesis and the deck.

13 · Cheat sheet

What	Number	Where
Participants	300; 62.3% aged 20–29; 44.0% medical students	Table 1
AI chatbots / search engines	78.0% (88.7% incl. named) / 56.3%; ChatGPT 59.3%	Figure 3
Delay care when reassured / never tell the doctor	34.7% / 31.7%	Figure 4
SHAI mean / CSS mean	12.3 / 28.9	Table 3
Reliability α	0.81 / 0.86	Table 3
Elevated health anxiety	19% (CI 15.0–23.8%); 32.7 / 19.0 / 7.3 / 0.7% at 15/18/22/29	Table 4
Invariance	CSS-12 scalar ✓; SHAI-14 metric ✓, scalar ×; no DIF	Table 5
Main correlation	r = 0.47 (0.38–0.55); partial 0.48; r^2 22%	Table 6, Figure 5
Facets	excess 0.46, distress 0.45, compulsion 0.25, reassurance 0.24	Table 6
Regression R^2	0.030 → 0.133 → 0.317; ΔR^2 18.4%; β 0.48; B 0.68	Table 9
Logistic	AOR 3.12 / 3.83 / 0.61; AUC 0.83 → 0.79	Table 10, Figure 8
Behaviour (robust)	CSS → never disclose AOR 0.57, q 0.012	Table 11
Mediation	via excess 0.16 (0.11–0.23); via distress 0.22, direct 0.01; reverse 0.15	Table 12, Figure 9
AI effect	+5.3 inclusive vs +1.5 strict (p 0.31); 32 named-not-ticked; 34 non-users	Table 9, slide 23
Item level	111/168 FDR; top ρ 0.43	Figure 11
Robustness	r 0.46–0.48 in every check	Table 13

> **Say:** *Health anxiety and cyberchondria travel together, r = 0.47; AI chatbots are the main source but have no reliable independent effect; and searching changes the consultation before it begins.*